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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 4.12

$$\begin{aligned}
 & \int \frac{\sqrt[4]{x} \sqrt[6]{x^7}}{\sqrt{x^3 \sqrt[12]{x^{11}}}} dx \\
 &= \int \frac{x^{\frac{1}{4}} x^{\frac{6}{7}}}{\sqrt{x^3 x^{\frac{11}{12}}}} dx \\
 &= \int \frac{x^{\frac{3}{12}} x^{\frac{14}{12}}}{\left(x^{\frac{36}{12}} x^{\frac{11}{12}}\right)^{\frac{1}{2}}} dx \\
 &= \int \frac{x^{\frac{17}{12}}}{\left(x^{\frac{47}{12}}\right)^{\frac{1}{2}}} dx \\
 &= \int \frac{x^{\frac{34}{24}}}{x^{\frac{47}{24}}} dx \\
 &= \int x^{\frac{34}{24}} x^{-\frac{47}{24}} dx \\
 &= \int x^{-\frac{13}{24}} dx \\
 &= \frac{x^{\frac{11}{24}}}{\frac{11}{24}} + c \\
 &= \frac{24}{11} \sqrt[24]{x^{11}} + c
 \end{aligned}$$

References